

Instructions for Using the Long Term Growth Model (LTGM) Spreadsheet (including poverty extension)

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Note: these instructions were originally designed for LTGM v4.4, but work for newer versions of the LTGM (with minor adjustments). To follow the instructions word-for-word, download the [Training version of LTGM](#) (v4.4).

Disclaimer: The views expressed here are the authors' and do not necessarily represent those of the World Bank. Example simulations are purely illustrative and do not necessarily reflect the economic situation in example countries (or the World Bank's view about economic prospects in that country).

Spreadsheet Overview:

- *InputDataA_GeneralAssumptions* and *InputDataB_ModelSpecAssumptions* are the only two worksheets ("tabs") where the user enters modelling choices:
 - Parameters and assumptions in *InputDataA* affect all Models (1-3).
 - Assumptions in *InputDataB* affect only a specific model.
- *GraphsB* plots **results** of each model (and model-specific assumptions from *InputDataB*).
- *GraphsA* plots all general assumptions in *InputDataA*.
- Tabs *Model1*, *Model1s*, *Model2*, *Model2s*, *Model3*, *Model3s* are where the calculations take place. *Model1*, *2*, *3* reflect the "baseline" and *Model1s*, *2s*, *3s* are the "scenario" to be analyzed.
 - Model 1: assume a path for the investment share of GDP (I/Y) → implied GDP growth.
 - Model 2: assume a path of GDP growth → required investment share of GDP (I/Y).
 - Model 3: assume a path for savings share of GDP (S/Y) (+ external sector) → implied GDP growth.
- **(NEW in LTGM v4.3)** There are three working age to total population ratios to choose from: 15-64, 15-60 and 20-64.
- **(NEW in LTGM v4.1)** The *DataSummary* tab summarizes data from different sources and time horizons, to give users an idea of trends, volatility and other measurement issues.
- **(NEW in LTGM v4.1)** Users can choose from different data sources/years for parameters/initial values using the dropdown menus next to each variable in *InputDataA* and *InputDataB*. Missing data from the default sources are interpolated based on income group medians, which triggers a red warning cell.
- **Color coding is important:** yellow indicates a cell the user can edit, red indicates one they shouldn't change, and grey indicates an inactive or summary cell.
- **(NEW in LTGM v4)** The model calculates how economic growth affects poverty rates.

Steps to run a simulation:

InputDataA_GeneralAssumptions Tab

- **Country, Year, and Parameters (top of InputDataA tab):** Choose the country (from the drop-down menu), initial year, and the three parameters required for the model (or use values pre-populated from data):
 - a. *initial K/Y* (should be between 1.5 and 3.5 for most countries)
 - b. *depreciation rate δ* (should be between 0.03 and 0.06 for most countries)
 - c. *labor share of income β* (should be between 0.4 and 0.8 for most countries)Users can also enter an initial value for real GDP per capita if they would like (eg using the most recent data).
- **Assumptions of future values of exogenous general growth drivers (left of InputDataA tab)** If on "automatic mode", the user can set an initial value for the growth driver, a target future value and a year to reach that value. The target and year achieved can vary across baseline and scenario. The model will then

draw a smooth path between initial and target values (and will be constant at the target value after that year). To set a constant growth rate just set the future growth rate equal to the current rate (the default), or set the target year as the initial year. The user can use the pre-populated initial value from a historical average, or choose an initial value (for example, using updated data) by editing the yellow cell.

- a. *Total factor productivity (A)* TFP is one of the most important drivers of growth. Reasonable values for long run annual TFP growth are between 0 and 0.02 (0%-2%) for most countries.¹
 - b. *Human capital growth (h)* is also important and growth is between 0 and 0.02 for most countries.
 - c. *Labor market participation rates by gender*. These are best kept constant for most countries at the initial value (where there is no effect on growth), the default. However, increased female labor force participation can be important for growth in countries which have low initial participation rates.
 - d. *Demographics*. Calculated by demographers based on trends in births, deaths and the population age structure. Best left on default (automatic) preloaded values for most users.
 - e. *External balance*. The external balance variables only affect growth in Model 3, but affect the savings rate and either CAB or debt in Model 1 and 2. The user first needs to choose whether to model the external sector via a future path of (i) external debt/foreign direct investment or (ii) the current account balance (CAB). (ii) is the simpler option, and with CAB/Y=0, Model 1 and 3 are identical. Depending on the option the user chooses from the drop-down menu, they need to select initial values, future targets and the target year in a similar way as other variables above.
- **Implied paths of exogenous growth drivers (right of InputDataA tab)** Check the implications of the choices made in (2) above for the future values of exogenous general growth drivers. If on automatic mode, no changes are allowed here. If one wants to choose a particular non-linear path for a variable, just change the drop-down menu to “manual” (previously was “automatic”) on the LHS. This will recolor the region on the right to yellow, indicating which cells can be edited manually year-by-year.
 - **A summary of the choices for future exogenous growth drivers are shown in tab *GraphsA*.**
 - **Effect of Growth on Poverty** At the bottom of InputDataA the user can set the parameters related to poverty, such as initial poverty headcount. The preloaded values are for extreme poverty (<\$1.90/day 2011PPP) from the most recent households survey in [PovcalNet](#), but the user can also input the poverty rate at the national (or other) poverty line.² How growth affects the poverty rate is governed by the Growth Elasticity of Poverty (GEP), for which there are two options:
 - a. The default is that income follows a *log-normal distribution* (generally a good approximation empirically) which automatically generates a relation between growth and poverty (and a GEP).
 - i. In this case, the user also has the option to select the path for future income inequality via the income Gini (or Shared Prosperity Premium). This is preloaded from the most recent household survey in PovcalNet or WDI but users can assume a different initial value, a future target or target year (or can choose a manual path for the Gini on the right). A lower Gini generally lowers poverty rates (other things equal) and increases the effect of growth on poverty
 - b. Alternatively, the user can input the Growth Elasticity of Poverty on “Manual” from the drop-down menu, and enter a path of GEP year-by-year in the yellow cells that appear on the right.

¹ Growth rates of many variables can be very volatile in the short term, but these variations even out over time. The user may choose different average over different horizons from the drop-down menu.

² Only the poverty headcount rate (not the value of the poverty line or specification in currency/period) affects simulations.

InputDataB ModelSpecAssumptions Tab

- **If using Model 1 (Input: Investment → Output: Growth)**, choose the initial investment share of GDP (or leave blank to use the pre-populated value), the target investment share and also the target year. The user can also choose a 20, 15, 10, 5-year average or the most recent available value for the investment-to-GDP ratio from a drop down menu. The assumed paths of investment are shown on the right. If the user prefers, choose manual (rather than automatic) and enter the investment share of GDP year-by-year in the yellow cells that appear on the RHS. Investment rates are usually between 10-40% of GDP, with investment rates of around 20%-25% being typical.
- **If using Model 2 (Input: Growth target → Output: Investment)**, the user first needs to choose a target from the drop-down menu: a [headline] GDP growth target, a GDP per capita (PC) target or a GDP per capita (PC) level target. The user chooses initial values, target values, target years and automatic / manual as above. Additionally, the user can choose a 20, 15, 10, 5-year average or the data point from the most recent available year from a drop-down menu for each variable. Unrealistic growth targets appear as unrealistically high required I/Y.
- **If using Model 3 (Input: Savings → Output: Growth)**, follow the same process for Model 1 in Step 5 above, but apply it to savings rather than investment. Savings rates and investment rates are roughly similar for most countries (the Feldstein-Horioka Puzzle): 10-40% with typical values of around 20%-25%. The difference in Model 3 is that growth depends on the external sector assumptions in *InputDataA* tab (CAB or External Debt/FDI), so the user of Model 3 needs to pay special attention to those assumptions. Also, the user can choose a 20, 15, 10, 5-year average or the most recent available data point from a drop-down menu.
- **Results are plotted in *GraphsB***, as well as model-specific inputs (which vary by model). Also included is the Marginal ICOR (number of ppts of investment/GDP required to boost growth by 1%) and the K/Y ratio. High marginal ICORs and K/Y ratios mean investment will be inefficient at producing growth.

Optional: Model 1, 1s, 2, 2s, 3 and 3s tabs – calculations at the heart of the model

A feature of the LTGM is the transparency and simplicity of the calculations of growth rates and required I/Y.

- “Summary of Inputs based on User's choice” at the top of each of these tabs which contain all the assumptions of inputs which are used in the model, and is linked to the RHS of *InputDataA* and *InputDataB*.
- “Endogenously Generated Variables” is the core of the LTGM - how growth and I/Y are calculated. These are the most important formulas, and reference only “Summary of Inputs”, or “Endogenously Generated Variables” from previous periods. Click in the formula bar to see which cells are referenced.
- “Memorandum Items” are extra variables that are calculated in the model, but do not affect growth or investment rates in current or future years.
- “Poverty-related calculations” are where the LTGM calculates the implications of growth for poverty rates. This includes both assumptions which are linked to the RHS of *InputDataA*, as well as the calculations of the poverty rates. For model 2 (with a poverty target), this part of the sheet calculated the growth rate required to achieve the poverty target.

On the next page are worked examples using the LTGM